

NITISH DUBEY

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EDUCATION

IIIT Naya Raipur

Aug 2023 – May 2027

B.Tech in Computer Science | CGPA: 8.1/10

- **Coursework:** DBMS, OS, Computer Networks, OOP, Blockchain Technology, Agentic AI, Cryptography

WORK EXPERIENCE

SAMSUNG

Nov 2025 – May 2026

R&D Intern | Remote

- Engineered an end-to-end multimodal RAG pipeline using LangChain, ChromaDB, and Flask to convert unstructured PDF manuals into structured step-by-step visual guides, reducing manual authoring effort.
- Automated large-scale document parsing with PyMuPDF and Vision LLMs to remove visual noise, generate semantic captions, and detect critical hardware objects for accurate guide creation.
- Built a semantic retrieval engine using HuggingFace embeddings and cosine similarity to map instructions with top relevant diagrams, improving instructional clarity and context accuracy.
- Developed a responsive full-stack web application using React, Vite, and Tailwind CSS, optimizing retrieval latency through precomputed cached embeddings and in-memory search.

PROJECTS

Rental Listing Platform | Wanderlust

Jan 2025 – Mar 2025

- Addressed fragmented rental booking workflows by building a full-stack platform using Node.js, Express.js, MongoDB, and EJS, enabling management of 30+ listings across multiple countries.
- Solved security and access control challenges by implementing Passport.js authentication, session management, owner-based authorization, and protected user workflows.
- Improved backend reliability by designing 12+ RESTful APIs for listings, reviews, and users with Joi validation, middleware sanitization, and scalable CRUD operations.
- Enhanced user experience and platform scalability by developing a responsive Bootstrap frontend and integrating Cloudinary with Multer for seamless booking flows and secure image uploads.

Fake News Verifier

Aug 2025 – Nov 2026

- Tackled online misinformation by architecting an AI-powered fact-checking platform using React, Node.js, and Express.js to verify user claims against real-time global news coverage.
- Improved evidence quality by engineering a retrieval pipeline with SerpAPI and a custom scoring algorithm that ranked sources by credibility, recency, and keyword relevance.
- Enhanced verification accuracy by integrating Llama 3.3 70B via Groq API to compare claims with retrieved evidence and generate evidence-backed fact-check reports.

SKILLS SUMMARY

- Languages: C++, Python, JavaScript (ES6+), SQL | Frontend: React.js, HTML5, CSS3, Tailwind CSS, Bootstrap, EJS | Backend: Node.js, Express.js, Flask, FastAPI, REST APIs
- Database & Cloud: MongoDB, Firebase, Supabase, Cloudinary, AWS (Basics) | AI/ML: scikit-learn, TensorFlow, Pandas, NumPy, NLP, LangChain, ChromaDB, OpenAI Agents SDK
- Tools & Platforms: Git, GitHub, Postman, Vercel, Render | Core Concepts: Authentication, CRUD Operations, API Integration, Data Structures & Algorithms

ACHIEVEMENTS & CERTIFICATIONS

- **Achievements:** Google Big Code – Shortlisted for Round 2 (Top 1500 students nationwide) | Algo University – Shortlisted for Round 2 (Top 2000 students nationwide) | Runner-up – E-Cell Hackathon
- **Certifications:** Certified Web Developer – Apna College